

Assignment On

‘ADVANCED DATA STRUCTURES AND ALGORITHMS’

(Assignment-4)

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Submitted to:
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Question:

Solve the Towers of Hanoi Problem Recursively. Also, Implement a Stack-Based Iterative Solution.

Code :

```
# -----  
  
# 1. RECURSIVE TOWERS OF HANOI  
# -----  
  
def recursive_hanoi(n, source, auxiliary, destination):  
    if n == 1:  
        print(f"Move disk 1 from {source} → {destination}")  
        return  
  
    recursive_hanoi(n - 1, source, destination, auxiliary)  
    print(f"Move disk {n} from {source} → {destination}")  
    recursive_hanoi(n - 1, auxiliary, source, destination)  
  
# -----  
  
# 2. ITERATIVE TOWERS OF HANOI USING STACKS  
# -----  
  
def move_disk(from_rod, to_rod, from_name, to_name):  
    disk = from_rod.pop()  
    to_rod.append(disk)
```

```
print(f"Move disk {disk} from {from_name} → {to_name}")
```

```
def legal_move(rod1, rod2, name1, name2):
```

```
    if not rod1: # rod1 empty → move from rod2 to rod1
```

```
        move_disk(rod2, rod1, name2, name1)
```

```
    elif not rod2: # rod2 empty → move from rod1 to rod2
```

```
        move_disk(rod1, rod2, name1, name2)
```

```
    elif rod1[-1] < rod2[-1]: # smaller disk moves
```

```
        move_disk(rod1, rod2, name1, name2)
```

```
    else:
```

```
        move_disk(rod2, rod1, name2, name1)
```

```
def iterative_hanoi(n):
```

```
    # Stacks (rods)
```

```
    A = list(range(n, 0, -1)) # Start rod (largest → smallest)
```

```
    B = []
```

```
    C = []
```

```
    total_moves = (2 ** n) - 1
```

```
    print("\nIterative Solution:")
```

```
    # Swap auxiliary and destination for even number of disks
```

```
    if n % 2 == 0:
```

```
        aux_name, dest_name = "C", "B"
```

```
    else:
```

```
        aux_name, dest_name = "B", "C"
```

```
    for move in range(1, total_moves + 1):
```

```
        if move % 3 == 1:
```

```
            legal_move(A, C, "A", "C")
```

```

elif move % 3 == 2:

    legal_move(A, B, "A", "B")

else:

    legal_move(B, C, "B", "C")

# -----

# MAIN PROGRAM (RUN BOTH SOLUTIONS)

# -----

if __name__ == "__main__":

    n = int(input("Enter number of disks: "))

    print("\n--- Recursive Towers of Hanoi ---")

    recursive_hanoi(n, "A", "B", "C")

    print("\n--- Iterative Towers of Hanoi (Using Stacks) ---")

    iterative_hanoi(n)

```

Output:

```

PS C:\Users\Madhu\OneDrive\Pictures\Desktop\ADSA\Assignment-4> & 'C:\Users\Madhu\AppData\Local\Programs\Python\Python314\python.exe' 'c:\Users\Madhu\.vscode\extensions\ms-python.debugpy-2025.16.0-win32-x64\bundled\libs\debugpy\launcher' '64035' '--' 'C:\Users\Madhu\OneDrive\Pictures\Desktop\ADSA\Assignment-4\task.py'
Enter number of disks: 3

--- Recursive Towers of Hanoi ---
Move disk 1 from A → C
Move disk 2 from A → B
Move disk 1 from C → B
Move disk 3 from A → C
Move disk 1 from B → A
Move disk 2 from B → C
Move disk 1 from A → C

--- Iterative Towers of Hanoi (Using Stacks) ---

Iterative Solution:
Move disk 1 from A → C
Move disk 2 from A → B
Move disk 1 from C → B
Move disk 3 from A → C
Move disk 1 from B → A
Move disk 2 from B → C
Move disk 1 from A → C
PS C:\Users\Madhu\OneDrive\Pictures\Desktop\ADSA\Assignment-4>

```